National University of Computer and Emerging Sciences 0024
Lahore Campus

Linear Algebra (MT1004) Sessional-II Exam

Date: April 9th 2026
Course Instructors
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Total Time (Hrs): 1
Total Marks: 40
Total Questions: 4

241-2606 BDS-4A

Roll No Section Student Signature

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Attempt all the questions.

CLO #2: Properties of vectors in 2-space, 3-space and n-space and recognize vector spaces and/or subspaces to compute their bases and its dimension

Q1. (a) [5]: Find **vector** and **parametric** equations of the plane in R^4 that passes through the point $(3, 0, -1, 2)$ and is parallel to both $(2, -4, 1, 5)$ and $(-8, 6, 0, 7)$ vectors.

$\frac{ax_0+bx_1+cx_2-d}{\sqrt{a^2+b^2+c^2}}$

(b) [5]: Find the distance between the planes $2x - y - z = 5$ and $-4x + 2y + 2z = 12$.

Q2. [10]: Show that the solution vectors of the following linear system are orthogonal to each row vector.

$$x_1 + 3x_2 - 2x_3 + 2x_5 = 0,$$
$$2x_1 + 6x_2 - 5x_3 - 2x_4 + 4x_5 - 3x_6 = 0,$$
$$5x_3 + 10x_4 + 15x_6 = 0,$$
$$2x_1 + 6x_2 + 8x_4 + 4x_5 + 18x_6 = 0.$$

Q3. [10]: Let **V** be the set of all ordered pairs of real numbers, and consider the following addition and scalar multiplication operations on $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$:

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1 + 1,\ u_2 + v_2 + 1), \quad k\mathbf{u} = (0, ku_2)$$

(i) Compute $\mathbf{u} + \mathbf{v}$ and $k\mathbf{u}$ for $\mathbf{u} = (-1, 2)$, $\mathbf{v} = (3, 4)$, and $k = 3$.
(ii) Find $\mathbf{0}$ and $-\mathbf{u}$.
(iii) Find vector space axioms that fail to hold (if exists).

Q4. [10]: If $A = \begin{bmatrix} 1 & 4 & 5 & 6 & 9 \\ 3 & -2 & 1 & 4 & -1 \\ -1 & 0 & -1 & -2 & -1 \\ 2 & 3 & 5 & 7 & 8 \end{bmatrix}$ then

(i) Find bases for the null space, column space and row space.
(ii) Find its rank and nullity.